

BrickAtlas Mechanism: Evaluating Constructible, Recoverable, and Functional Brick Systems

Development Manuscript

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Figure 1. **Six original articulated benchmark objects.** The rover, gantry crane, bascule gate, observatory, tiltrotor, and polar station contain 542 visible brick-system parts, 36 rigid modules, and 33 fixed, revolute, or prismatic joints. OMR/LDraw models informed the target level of visual coherence but are not benchmark cases.

Abstract

Brick-assembly evaluation has advanced in generation, spatial reasoning, and final-state validity, but a valid final object does not certify a constructible, recoverable, or functional process. We introduce BrickAtlas Mechanism-1, a diagnostic benchmark built around six original articulated objects and eight task families: prefix dynamics, continuous service access, causal fault recovery, finite-inventory substitution, dynamic impulse robustness, functional kinematics, active inspection, and multi-objective design. Unlike our retired random-growth layer, all publication-facing objects are coherent systems with wheels, axles, gears, hinges, sliding carriages, payloads, and service bays. Rapier 3D supplies deterministic rigid-body simulation, fixed/revolute/prismatic joints, motor actuation, and swept-volume shape casts. The release contains 48 paired tasks, 72 audited renders, complete machine-readable inputs and oracles, and

a per-case browser. Under the same one-image minimal harness, GPT-4.1 mini solves 28/48 cases and Gemini 2.5 Flash 31/48; a local Qwen3-0.6B text baseline solves 3/48. Prefix and finite-stock questions are substantially easier than continuous access, exact repair, functional outcomes, and complete Pareto sets. The six public sources are development data, and simulator evidence is not a calibrated claim about LEGO clutch force or robot execution.

1. Introduction

Recent brick systems generate plausible objects, infer assembly steps, or answer spatial questions. Yet many evaluations still terminate at a final structure or a single next action. Real construction requires more: every prefix must survive gravity, a part or tool must reach its target, failures must be diagnosed and repaired locally, mechanisms must move as

intended, and an agent must ask for information when the state is ambiguous.

These requirements are coupled. A visually convincing crane can have an inaccessible trolley; a correct drawbridge geometry can have a reversed hinge; a robust plan can become invalid after one missing brace; and a cheap replacement can violate stiffness requirements. Aggregating these failures into final IoU, validity, or a language-model judge hides what a system actually accomplished.

BrickAtlas Mechanism-1 asks whether a model can reason about *constructible processes* on shared, functional objects. Figure 1 shows the complete source set and Fig. 2 the eight evaluation families. We intentionally do not import OMR geometry as cases: those models are retained only as design-complexity references. The previous 5,120 random-growth objects are likewise retired to regression testing.

Our contributions are:

1. six original articulated brick-system designs with explicit rigid modules, collision geometry, joints, service interfaces, inventories, and hidden-state alternatives;
2. eight gap-driven tasks with deterministic physics, collision, optimization, and information-gain oracles;
3. a fully inspectable case browser exposing every question, capability, input, oracle, and model answer; and
4. a paired development study of two VLMs and one local text model, with raw outputs, costs, and source-grouped reporting.

2. Related Work

Generation and final validity. BrickGPT [8] and BrickNet [4] generate brick structures using physical checks or typed connection graphs. BrECS [1] studies budget-aware sequential assembly. These works establish that stable generation and resource constraints are active research areas; our target is not another text-to-brick generator, but process-level diagnosis after a design exists.

Visual and procedural reasoning. LEGO-Puzzles [9] and PhyBlock [6] cover spatial, ordering, diagnostic, and counterfactual questions. MEPNet [11], Neural Assembler [12], and TreeSBA [2] infer poses or sequences from manuals and views. Co-builder [3] studies assembly-state detection. Mechanism-1 complements these benchmarks with continuous shape casts, dynamic thresholds, explicit joints, minimal recovery, and costed clarification.

Interactive and embodied assembly. Break and Make [10] uses interactive disassembly and reconstruction. WorkBenchMark [7] includes robot/VLA execution, while BC-Bench [5] evaluates selection and pose. We do not claim robot realism. Rapier provides deterministic digital

Object	Parts	Modules	Joints
Orbital Service Rover	73	10	9
Harbor Container Crane	126	5	5
Bascule Canal Gate	142	6	7
Adaptive Radio Observatory	92	4	3
Wildfire Tiltrotor	43	6	5
Polar Research Station	66	5	4

Table 1. Mechanism-1 source objects. “Parts” are visible brick-system elements; physics runs on compound rigid modules.

rigid-body evidence; grasp, tolerance, clutch force, and hardware transfer remain separate questions.

3. Objects and Representation

3.1. Original Mechanism Sources

The six objects span planetary maintenance, load handling, movable infrastructure, scientific instrumentation, aerial response, and remote infrastructure (Table 1). Each is authored for this benchmark and combines visually coherent subassemblies rather than randomly grown bricks.

A visible part has shape, size, local pose, material color, and module membership. The vocabulary includes bricks, plates, beams, panels, slopes, arches, windows, cylinders, wheels, axles, gears, and spheres. A module has a world pose, mass, role, and anchored/dynamic state. Joints are fixed, revololute, prismatic, or spring descriptors with local anchors, axes, limits, and optional actuation.

OMR/LDraw examples are not included in the task manifest, source groups, scores, or model inputs. They only guided an internal design review of object coherence and subassembly granularity.

3.2. Physics and Collision Oracle

We use Rapier 3D v0.20 at 120 Hz, 12 solver iterations, and two internal PGS iterations. Compound colliders approximate the visual parts; slopes and arches use conservative cuboid envelopes. Prefix evaluation rebuilds each partial assembly, enables available joints, runs one simulated second, and records the first module displacement above 0.35 units.

Accessibility uses a continuous cuboid shape cast from start pose to service interface. Dynamic robustness applies an off-center impulse and measures

$$d_{\max} = \max_t \max(\|\mathbf{x}_t - \mathbf{x}_0\|, r\theta_t), \quad (1)$$

where r is the target module radius and θ_t its quaternion rotation from the settled pose. Functional tasks motor a named joint and compare measured motion with a declared fraction of the target. All thresholds and engine settings are public and frozen.

4. Eight Gap-Driven Tasks

Prefix dynamics and access. These tasks operationalize two widely noted gaps: final validity does not imply stable prefixes, and a target pose does not imply an executable insertion. The candidate order is not silently repaired. Shape casts return every feasible path rather than one private reference.

Recovery and resources. Fault cases remove one structural or service joint, expose the simulated child displacement and tilt, and require a complete minimum repair program. Inventory cases force selection among a costly stiff beam, paired trusses, a light cable, and an ineffective fairing under explicit piece, cost, and stiffness limits.

Dynamics and function. Robustness is a threshold task over measured impulses, not a static “stable” label. Functional cases deliberately include successful and failed actuation outcomes caused by load, joint limits, or interference. Thus appearance alone cannot solve the task.

Inspection and engineering trade-offs. Active inspection scores expected information gain per unit cost over four hidden worlds: nominal, reversed, missing-brace, and jammed. Query costs vary by source, preventing one fixed query from succeeding. The design task requires the complete Pareto set under lower cost, lower mass, and higher stiffness.

5. Evaluation Protocol

Each task supplies one rendered view and structured evidence under a fixed minimal harness. Models return one JSON object; unknown fields do not repair missing required values. Missing, malformed, truncated, and duplicate submissions remain in the denominator. Exact success is reported separately for all eight families, with the source object as the statistical cluster.

The deterministic tracks are: **physics oracle**, which executes Rapier dynamics or shape casts; **constraint oracle**, which enumerates inventory/Pareto answers; and **information oracle**, which computes finite-world query utility. Model outputs never receive oracle feedback.

We test GPT-4.1 mini and Gemini 2.5 Flash with the same image, structured input, temperature zero, 2,200-token cap, no retry, and no provider fallback. Qwen3-0.6B is a local text-only lower baseline. All 96 paid calls cost \$0.069427; cumulative project spend remains below the pre-existing \$4.50 cap.

6. Results

GPT-4.1 mini solves 28/48 cases, Gemini 31/48, and local Qwen3-0.6B 3/48. Both VLMs solve all finite-inventory cases, while prefix accuracy is 5/6 and 6/6. This does not make those families redundant: the valid order and numeric limits are explicitly represented, and the six-source scale is small.

The largest gaps are more diagnostic. GPT solves 3/6 access cases and no exact repair case; Gemini solves 3/6 access and 3/6 repair. Functional outcomes are 3/6 for both because some actuators fail in simulation despite valid-looking geometry. GPT obtains 5/6 dynamic thresholds while Gemini obtains 2/6. Complete Pareto sets remain difficult (2/6 and 4/6). These failures persist despite perfect output formatting, so they are not explained by JSON parsing.

The task families also avoid one common answer template. Feasible path sets vary across at least four patterns; safe impulse thresholds vary across at least four values; active-inspection optima cover marker, brace, and probe queries; and functional labels include both outcomes. The local model’s 3/48 result further rejects a broad format-only shortcut.

7. Limitations

Mechanism-1 contains only six public authored sources. The objects were used during task design, so they are development data rather than an independent confirmation set. Visual parts are LEGO-like primitives, not a claim of compatibility with every commercial element.

Rapier validates rigid-body, joint, motor, and collision consequences under declared parameters. It does not calibrate ABS elasticity, stud clutch, manufacturing tolerance, wear, gripper contact, or human ergonomics. Slope and arch physics use conservative collision envelopes. Real-brick and robot transfer require separate experiments.

The model comparison is descriptive: six source clusters cannot support fine-grained ranking or broad population claims. Human difficulty, independent object design, additional model families, multi-seed sampling, and real hardware remain outstanding. The complete browser and Chinese casebook expose every task and answer so these limitations can be audited rather than hidden behind an aggregate score.

8. Conclusion

BrickAtlas Mechanism shifts brick evaluation from final appearance and axis-aligned validity to process, intervention, and function. Original articulated objects, external rigid-body simulation, continuous shape casts, exact repair programs, active inspection, and Pareto scoring produce failures that earlier recoloring and height-sorting tasks could

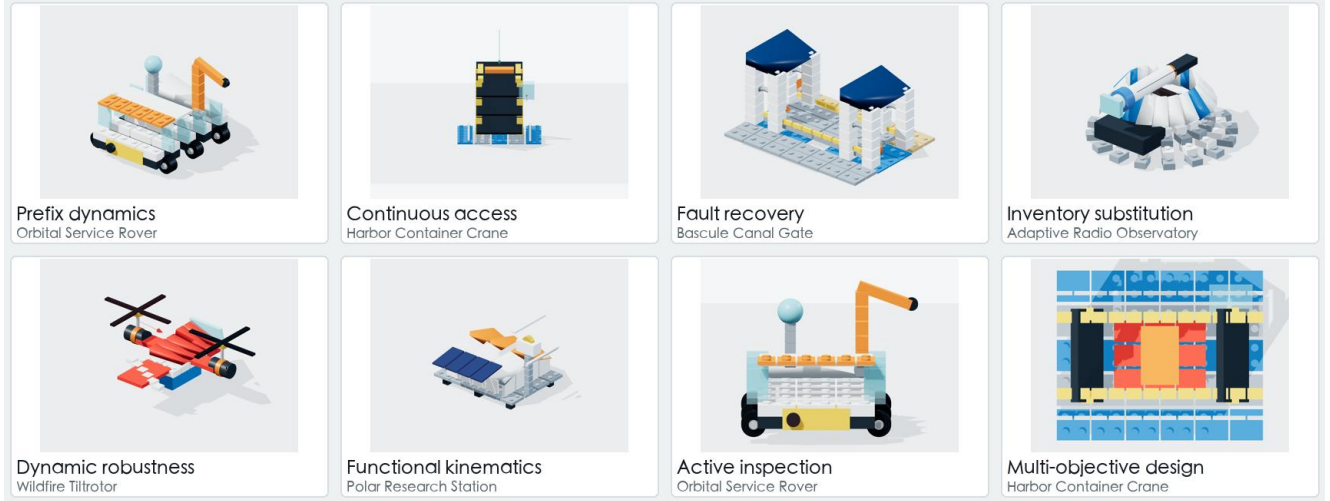


Figure 2. **The complete task taxonomy.** Each family contains one case for every source object. Dark/red highlighting identifies the queried module; access views overlay green feasible and red colliding paths; robustness views show the impulse direction.

Family	Public evidence	Output	Exact success
Prefix dynamics	two module orders, joints, masses	valid plan, first failed prefix	both fields correct
Insertion access	service target, three swept paths	all accessible path IDs	exact feasible set
Fault recovery	displacement/tilt, candidate joints	fault joint, ordered actions	diagnosis and sequence exact
Inventory substitution	alternatives, stock/cost/stiffness limits	replacement ID	all limits satisfied
Dynamic robustness	impulse ladder and displacement limits	largest safe impulse	exact threshold
Functional kinematics	joint type, axis, target motion	joint ID, pass/fail	exact simulated outcome
Active inspection	finite worlds, query returns/costs	next query ID	max information/cost
Multi-objective design	cost, mass, stiffness alternatives	all Pareto IDs	exact non-dominated set

Table 2. Every family has six paired cases and a deterministic evaluator. No LLM judge defines correctness.

not reveal. The first model study confirms substantial head-room in accessibility, recovery, functional prediction, dynamic thresholds, and multi-objective reasoning, while preserving explicit boundaries between simulation evidence and real-world mechanics.

Model	Prefix	Access	Repair	Stock	Dynamics	Function	Inspect	Pareto
Qwen3-0.6B (text)	1/6	0/6	0/6	0/6	0/6	0/6	2/6	0/6
GPT-4.1 mini	5/6	3/6	0/6	6/6	5/6	3/6	4/6	2/6
Gemini 2.5 Flash	6/6	3/6	3/6	6/6	2/6	3/6	4/6	4/6

Table 3. Exact successes out of six source objects per family. Qwen is text-only; GPT and Gemini receive one task render plus identical structured evidence. This is a development comparison, not a stable leaderboard.

References

- [1] Seokjun Ahn, Jungtaek Kim, Minsu Cho, and Jaesik Park. Budget-aware sequential brick assembly with efficient constraint satisfaction. *Transactions on Machine Learning Research*, 2024. [2](#)
- [2] Mengqi Guo, Chen Li, Yuyang Zhao, and Gim Hee Lee. TreeSBA: Tree-transformer for self-supervised sequential brick assembly. In *European Conference on Computer Vision (ECCV)*, 2024. [2](#)
- [3] Haochen Huang, Yue Su, Xin Sun, Moonisa Ahsan, Mohammad Aliannejadi, Irene Viola, Zhaochun Ren, Chuang Yu, Aneta Lisowska, Artem Belopolsky, Koen Hindriks, Pablo Cesar, Junxiao Wang, and Jiahuan Pei. LEGO Co-builder: Exploring fine-grained vision-language modeling for multimodal assembly assistants. *arXiv preprint arXiv:2507.05515*, 2025. Version 4 revised July 2026. [2](#)
- [4] Peter Kulits and Cordelia Schmid. BrickNet: Graph-backed generative brick assembly. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 39252–39261, 2026. [2](#)
- [5] Jiateng Liu, Bingxuan Li, Zhenhailong Wang, Rushi Wang, Kaiwen Hong, Cheng Qian, Jiayu Liu, Denghui Zhang, Katherine Driggs-Campbell, Manling Li, and Heng Ji. Brick-composer: MLLMs construct everything from building blocks. *arXiv preprint arXiv:2606.05445*, 2026. [2](#)
- [6] Liang Ma, Jiajun Wen, Min Lin, Rongtao Xu, Xiwen Liang, Bingqian Lin, Jun Ma, Yongxin Wang, Ziming Wei, Haokun Lin, Mingfei Han, Meng Cao, Bokui Chen, Ivan Laptev, and Xiaodan Liang. PhyBlock: A progressive benchmark for physical understanding and planning via 3d block assembly. In *Advances in Neural Information Processing Systems*, 2025. [2](#)
- [7] Wenbo Ma, Daniel Swoboda, Matteo Tschesche, and Till Hofmann. WorkBenchMark: A LEGO-based assembly benchmark with an assembly-by-disassembly baseline for the smart manufacturing league. *arXiv preprint arXiv:2606.19358*, 2026. [2](#)
- [8] Ava Pun, Kangle Deng, Ruixuan Liu, Deva Ramanan, Changliu Liu, and Jun-Yan Zhu. Generating physically stable and buildable brick structures from text. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14798–14809, 2025. [2](#)
- [9] Kexian Tang, Junyao Gao, Yanhong Zeng, Haodong Duan, Yanan Sun, Zhening Xing, Wenran Liu, Kaifeng Lyu, and Kai Chen. LEGO-Puzzles: How good are MLLMs at multi-step spatial reasoning? *arXiv preprint arXiv:2503.19990*, 2025. Revised public submission includes a 1–8-step planning set; accessed 2026-09-15. [2](#)
- [10] Aaron Walsman, Muru Zhang, Klemen Kotar, Karthik Desingh, Ali Farhadi, and Dieter Fox. Break and make: Interactive structural understanding using LEGO bricks. In *European Conference on Computer Vision (ECCV)*, 2022. [2](#)
- [11] Ruocheng Wang, Yunzhi Zhang, Jiayuan Mao, Chin-Yi Cheng, and Jiajun Wu. Translating a visual LEGO manual to a machine-executable plan. In *European Conference on Computer Vision (ECCV)*, 2022. [2](#)
- [12] Hongyu Yan and Yadong Mu. Neural assembler: Learning to generate fine-grained robotic assembly instructions from multi-view images. *arXiv preprint arXiv:2404.16423*, 2024. [2](#)